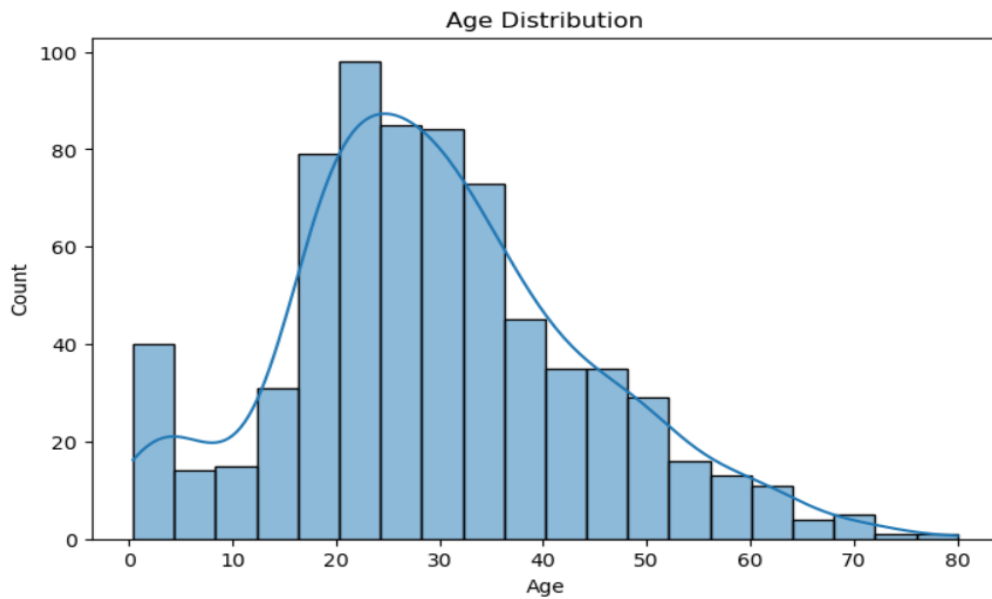


Exploratory Data Analysis (EDA) Report on Titanic Dataset

Objective: Analyze the Titanic dataset and identify patterns affecting passenger survival.

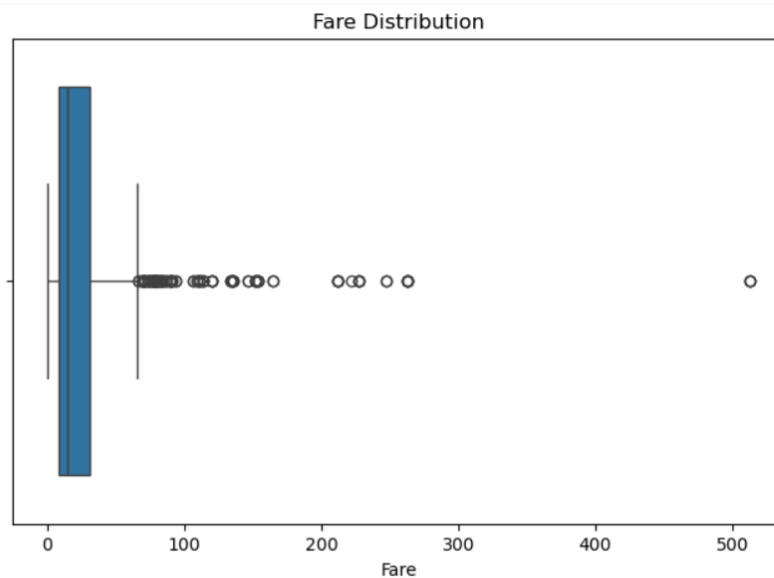
Tools: Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook (task_5.ipynb)

Age Distribution



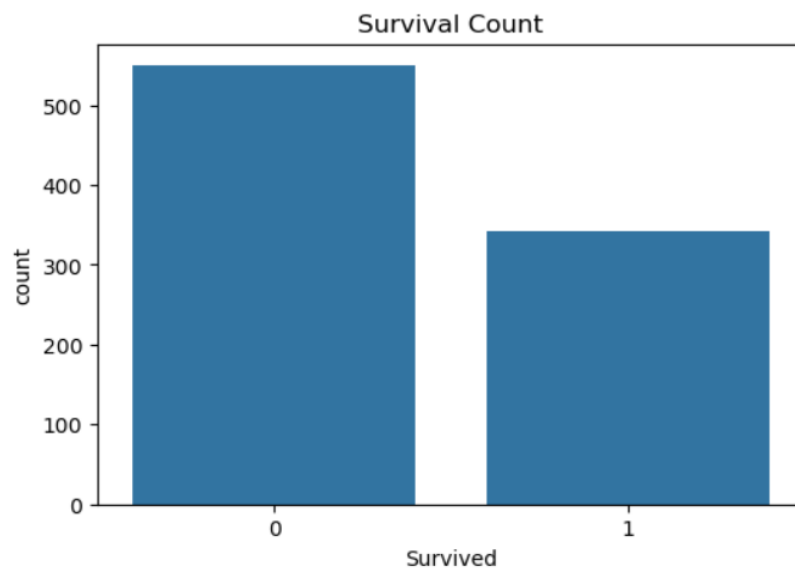
Most passengers were between 20 and 40 years old. The distribution is slightly right-skewed.

Fare Distribution



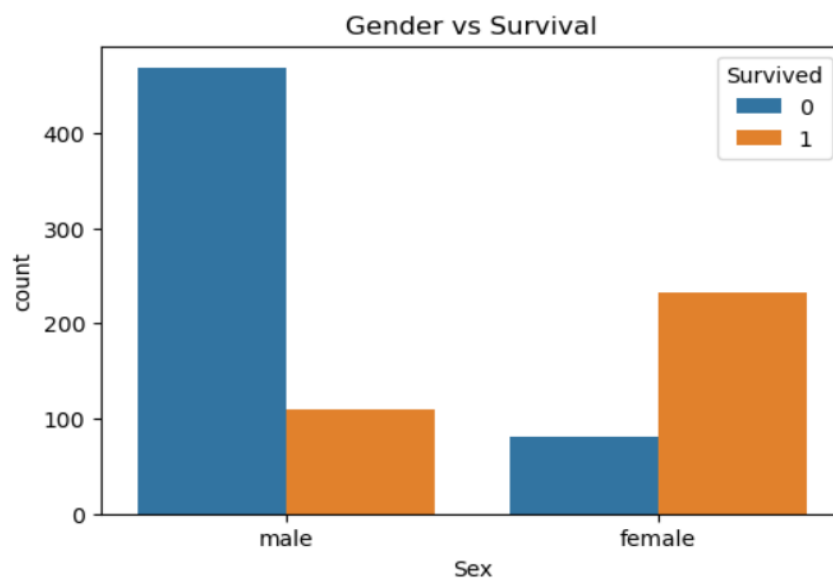
Fare contains significant outliers and is highly skewed.

Survival Count



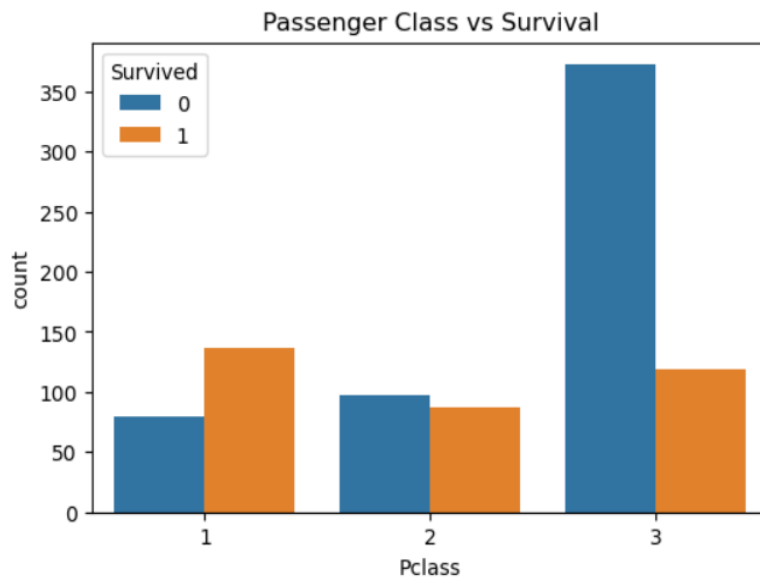
More passengers died than survived. Survival rate is approximately 38%.

Gender vs Survival



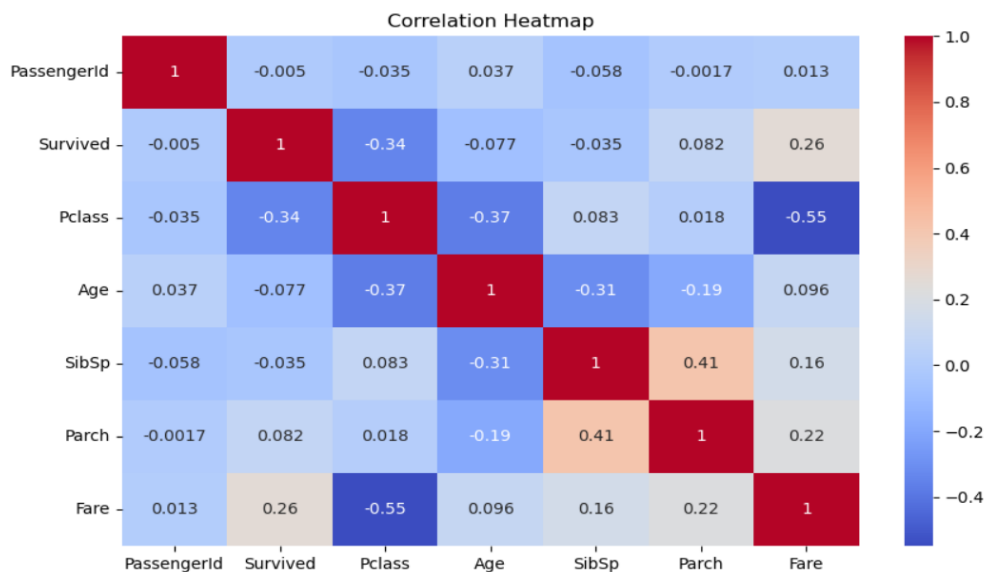
Female passengers had a much higher survival rate than males.

Passenger Class vs Survival



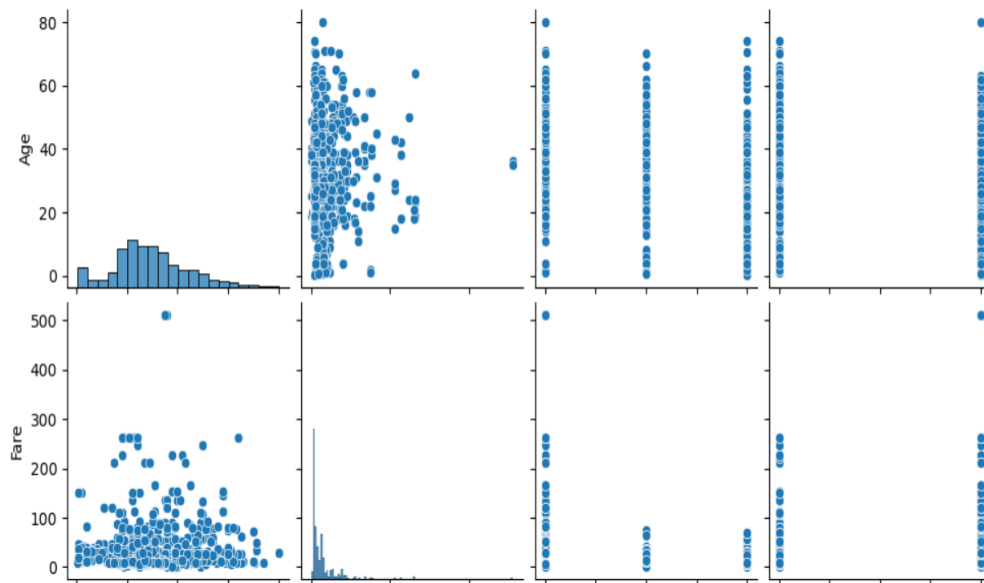
First-class passengers survived more frequently, while third-class passengers had the highest mortality.

Correlation Heatmap



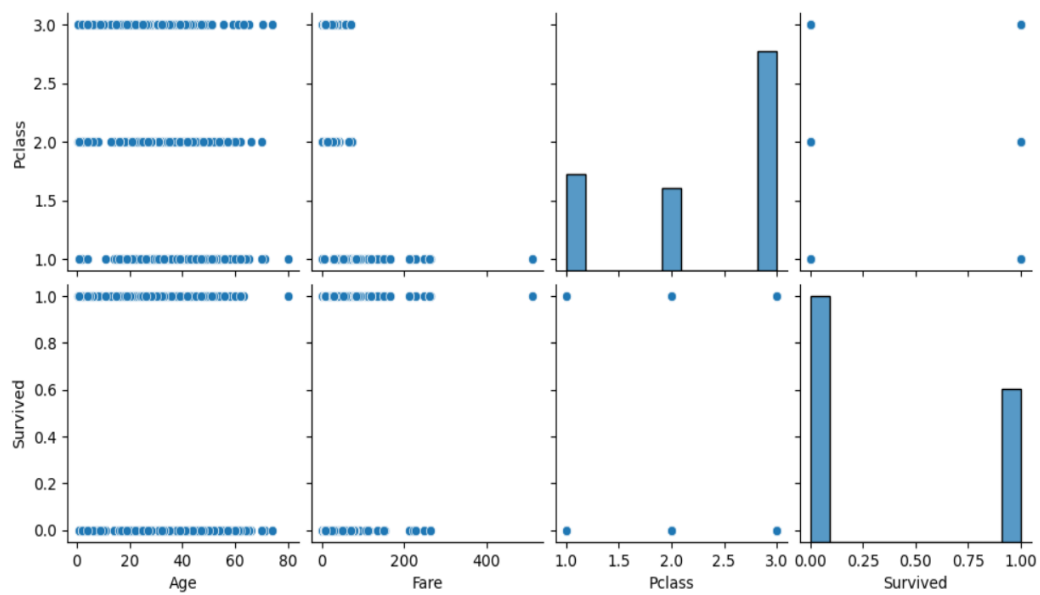
Passenger class and fare show meaningful relationships with survival.

Pairplot Analysis



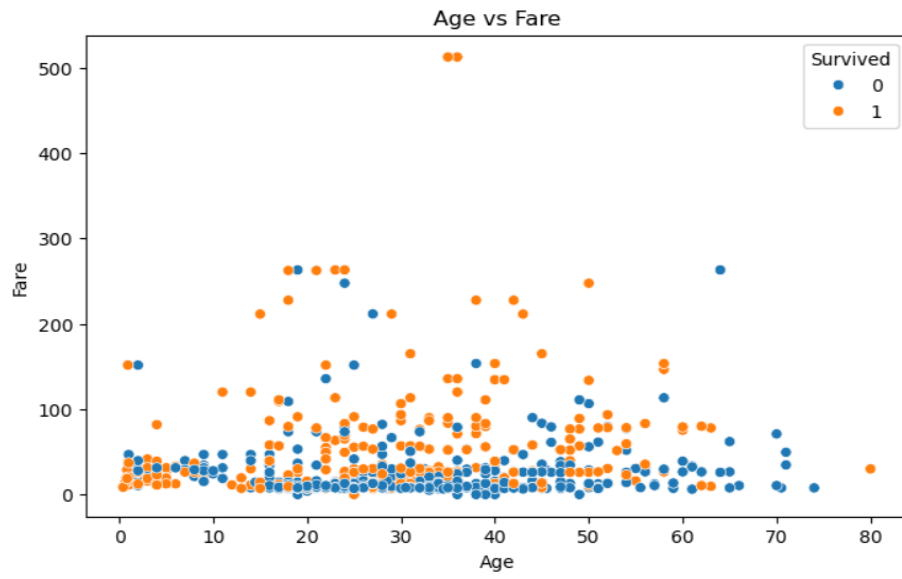
Fare and passenger class show stronger relationships with survival than age.

Additional Pairplot View



Relationships among Age, Fare, Pclass, and Survival are visualized.

Age vs Fare Scatter Plot



Higher fare passengers generally had better survival chances across age groups.

Conclusion

Gender, passenger class, and fare were the strongest factors influencing survival. Female and first-class passengers had significantly better survival outcomes.